Lesson 1

Need To Visualise Data

Why to Visualise data?

I. 65% of humans are better Visual Learners.Vision is a powerful driver for learning and receiving information.

II. We retain 80% of information seen as compared to 20% of information read.

III. Visual perception is the fastest amongst all senses at 13 ms.

IV. We will forget textual information easily as compared to visual information.

Helps In ?

1. Decision Making
2. Uncovering Patterns and Trends
3. Presenting Arguments/Information and/or Telling a story.

There are cases when the descriptive statistics for different sets of data are the same but that does not mean that the dataset is the same. There could still be fundamental differences in the Nature of the Dataset and plotpoints.

When we plot the datasets, we can see they have different distributions.

Reasons to create visualization can be :

• Visualization compresses large volume of data into easy-to-understand visuals
• Enables business intelligence
• Effective for exploration of the data as well communication of insights from the data
• Discovering answers to questions from the underlying data
• Enabling data aid decision making
• Understanding the data in a context
• Finding hidden patterns in the data
• Presenting an argument or telling a story
• Inspiring or persuading with data story

How Visualization Helps Communicate Complex Information

It Helps in Communicating with Clarity. It Simplifies complex data for all audiences using pre-attentive processing , Reduces cognitive load with intuitive designs, Engages viewers with memorable, actionable insights.

Gestalt principles is a collection of visual perception principles that aid us in understanding how human perception works.
These principles explain how humans tend to interpret structure, logic, and pattern around them.

Purpose : Derived from psychological Research to guide better visual design.
Basis : Explains how we perceive structure, logic , and patterns in visuals.
Application : Helps create clear, effective visualisations by distinguishing the subject from its background.

A. Principle of Figure and Ground

1. This states that when we look at a visual, our brain instinctively distinguishes the objects as either they are in foreground or background.
2. Helps to Focus on main elements when you develop charts and put those elements in the foreground.
3. Example : Rubine vase faces.

B. Principle of similarity

1. This states if we group objects together then they appear to be similar.
2. Several Visual elements used to show the similarity among different groups such as shape, size or color.

C. Principle of Focal Point :

1. This states that whatever is different or distinct will be the focus point that catches the viewer's attention.

D. Principle of Continuity :

1. This states that the human brain is more likely to perceive a continuous, smooth path when they visualize a line or a curve, regardless of their shape and color.

E. Principle of Common Region :

1. This states that things that belong to the same closed region are perceived as a group.
2. By creating a boundary, we can create a perception that within that closed region things function similarly.

F. Principle of closure :

1. This states that the human brain tends to find a single known pattern whenever it looks at a complicated, incomplete arrangement of visual elements.

These Principles play an important role while building effective visualisation. One can incorporate these principles in their visualisations to grab the audience's attention for long and immediately.